Day 1 Notes: Introduction to Topological Geometry and the h-Cobordism Theorem

Geometry and Topology Summer School

Contents

1 Introduction to Topological Geometry

1.1 Manifolds and the main problems

A manifold may be studied in several categories, most notably the *topological* and *smooth* (C^∞) categories. Some of the central questions in topological geometry are:

- **Classification**: how can manifolds be classified up to homeomorphism or diffeomorphism?
- **Maps**: which maps between manifolds are meaningful, and when can a homotopy equivalence be improved to a homeomorphism or diffeomorphism?
- **Symmetries**: how do group actions on manifolds reflect their topology and geometry?

- **General structures**: for example, smooth or complex structures on a fixed topological manifold.

  A rough slogan for the subject is

$$\boxed{\text{Homotopy theory} \quad \longrightarrow \quad \text{topological manifolds} \quad \longrightarrow \quad C^\infty\text{-manifolds}}$$

with geometric topology providing a bridge between these viewpoints, especially in high dimensions.

Question 1

Given a finite complex K, does there exist a closed manifold M^n such that

$$M \simeq K?$$

If so, how many such manifolds are there?

Question 2

Given a topological manifold M^n, does M admit a C^∞-structure? If it does, how many smooth structures can it carry?

1.2 Homotopy equivalence does not imply homeomorphism

A natural but false rigidity principle would assert that every homotopy equivalence between closed manifolds is homotopic to a homeomorphism. Lens spaces give classical counterexamples to this principle.

1.2.1 Construction of $L^3(p,q)$

Write

$$S^3 = \left\{(z_1, z_2) \in \mathbb{C}^2 : \ |z_1|^2 + |z_2|^2 = 1\right\}.$$

Let $\mathbb{Z}_p$ act on S^3. If

$$\zeta = e^{2\pi i/p}$$

is a generator and $(p,q) = 1$, define

$$\zeta \cdot (z_1, z_2) = (\zeta z_1, \zeta^q z_2) = \left(e^{2\pi i/p} z_1, e^{2\pi i q/p} z_2\right).$$

Fact 1.1. The above $\mathbb{Z}_p$-action on S^3 is free.

Indeed, if $\zeta^m \cdot (z_1, z_2) = (z_1, z_2)$, then the coprimality condition forces $m \equiv 0 \pmod p$.

Fact 1.2. For a finite group acting freely on a Hausdorff space, the action is automatically properly discontinuous.

Therefore, by the quotient-manifold theorem,

$$L^3(p,q) := S^3/\mathbb{Z}_p$$

is a smooth 3-manifold.

Thus $S^3 \to L^3(p,q)$ is the universal covering map.

Reserved illustration

Lens-space quotient: the universal covering $S^3 \to L^3(p,q) = S^3/\mathbb{Z}_p$.

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Figure 1: Lens-space quotient: the universal covering $S^3 \to L^3(p,q) = S^3/\mathbb{Z}_p$.

1.2.2 Basic classification facts for lens spaces

For $(p, q_i) = 1$ one has

$$L^3(p,q_1) \simeq L^3(p,q_2) \iff q_1 q_2 \equiv \pm a^2 \pmod p \quad \text{for some } a \in \mathbb{Z},$$
$$L^3(p,q_1) \cong_{\text{homeo}} L^3(p,q_2) \iff q_2 \equiv \pm q_1^{\pm 1} \pmod p,$$

where the second criterion is for homeomorphism without a prescribed orientation. In particular,

$$L^3(7,1) \simeq L^3(7,2), \qquad L^3(7,1) \not\cong_{\text{homeo}} L^3(7,2).$$

Reidemeister torsion is the classical invariant that distinguishes such lens spaces beyond homotopy type.

1.3 The simply connected case and Novikov's theorem

Homotopy equivalence is still too weak in the simply connected category. There are homotopy equivalences

$$f : M \longrightarrow N$$

between closed simply connected smooth manifolds for which

$$p_1(M) \otimes 1 \neq f^* p_1(N) \otimes 1 \qquad \text{in } H^4(M;\mathbb{Q}).$$

Such manifolds cannot be homeomorphic through a map homotopic to f.

Theorem 1.3 (Novikov)**.** For a homeomorphism $h : M \to N$ between smooth manifolds,

$$h^* p_i(N) = p_i(M) \qquad \text{in } H^{4i}(M;\mathbb{Q})$$

for every i. Thus rational Pontryagin classes depend only on the underlying topological manifold.

Hence rational Pontryagin classes provide obstructions to improving a homotopy equivalence to a homeomorphism.

1.4 The Borel conjecture

Definition 1.4. A closed manifold M is *aspherical* if

$$\pi_i(M) = 0 \qquad \text{for all } i \geq 2.$$

Equivalently, its universal cover $\widetilde{M}$ is contractible.

Thus an aspherical manifold is a model for the Eilenberg–Mac Lane space

$$K(\pi_1(M), 1).$$

Conjecture 1.5 (Borel)**.** Let M^n be a closed aspherical manifold. Then every homotopy equivalence

$$f : N^n \longrightarrow M^n$$

is homotopic to a homeomorphism.

Heuristically, this says that a closed aspherical manifold should be topologically determined by its fundamental group.

1.4.1 Examples of aspherical and non-aspherical spaces

1. S^1 is aspherical because $\widetilde{S^1} \cong \mathbb{R}$.
2. S^2 is not aspherical because $\widetilde{S^2} = S^2$.
3. T^2 is aspherical because $\widetilde{T^2} \cong \mathbb{R}^2$.

For a closed surface Σ_g of genus $g \geq 2$, uniformization gives

$$\Sigma_g \cong \mathbb{H}^2/\Gamma$$

for a discrete group Γ acting on the hyperbolic plane.

Reserved illustration

Uniformization picture for an aspherical surface: $\Sigma_g \cong \mathbb{H}^2/\Gamma$.

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Figure 2: Uniformization picture for an aspherical surface: $\Sigma_g \cong \mathbb{H}^2/\Gamma$.

1.5 Mostow rigidity

Theorem 1.6 (Mostow rigidity)**.** Let M^n and N^n be closed hyperbolic manifolds of dimension $n \geq 3$ (with sectional curvature -1). Then every homotopy equivalence

$$f : M \longrightarrow N$$

is homotopic to an isometry.

This is a particularly strong form of rigidity: the homotopy type determines the hyperbolic metric up to isometry.

1.6 Nonpositive curvature and Cartan–Hadamard

Let (M^n, g) be a closed Riemannian manifold with

$$\sec(M, g) \leq 0.$$

Then M is aspherical. The relevant form of the Cartan–Hadamard theorem is:

Theorem 1.7 (Cartan–Hadamard)**.** If $\widetilde{M}$ is complete, simply connected, and has nonpositive sectional curvature, then for every $\widetilde{x} \in \widetilde{M}$ the exponential map

$$\exp_{\widetilde{x}} : T_{\widetilde{x}}\widetilde{M} \longrightarrow \widetilde{M}$$

is a diffeomorphism.

Hence

$$\widetilde{M} \cong T_{\widetilde{x}}\widetilde{M} \cong \mathbb{R}^n,$$

so M is aspherical.

Reserved illustration

Cartan–Hadamard picture: $T_{\widetilde{x}}\widetilde{M} \cong \mathbb{R}^n$ and the exponential map $\exp_{\widetilde{x}}$.

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Figure 3: Cartan–Hadamard picture: $T_{\widetilde{x}}\widetilde{M} \cong \mathbb{R}^n$ and the exponential map $\exp_{\widetilde{x}}$.

1.7 Aspherical manifolds from Lie groups

Let

$$G = SL_n(\mathbb{R}), \qquad K = SO(n).$$

The symmetric space

$$X = G/K$$

can be identified with the space of positive-definite symmetric matrices of determinant 1. In particular, X is contractible and is diffeomorphic to a Euclidean space of dimension

$$\frac{n(n+1)}{2} - 1.$$

If $\Gamma \subset G$ is a discrete torsion-free subgroup, then the quotient

$$\Gamma\backslash X$$

is an aspherical manifold.

For an integer $m \geq 3$, the principal congruence subgroup

$$\Gamma(m) = \ker\big(SL_n(\mathbb{Z}) \longrightarrow SL_n(\mathbb{Z}/m\mathbb{Z})\big)$$

is torsion-free. Hence

$$\Gamma(m)\backslash SL_n(\mathbb{R})/SO(n)$$

is a standard arithmetic example of an aspherical manifold. Such locally symmetric manifolds form an important family in the study of topological rigidity.

1.8 The generalized Poincaré conjecture

Theorem 1.8 (Generalized Poincaré theorem, topological form)**.** *If a closed manifold M^n is homotopy equivalent to S^n, then*

$$M^n \cong_{\text{homeo}} S^n.$$

One may then ask separately whether M is diffeomorphic to S^n.

If $M \simeq S^n$, then

$$\pi_1(M) = 0, \qquad H_*(M) \cong H_*(S^n).$$

The smooth h-cobordism argument developed below applies directly in dimensions $n \geq 6$.

1.8.1 Bad news: fundamental groups can be arbitrary

Let

$$\Gamma = \langle a_1, \ldots, a_m \mid r_1, \ldots, r_k \rangle$$

be a finitely presented group. For every $n \geq 4$, there exists a closed smooth n-manifold M^n with

$$\pi_1(M) \cong \Gamma.$$

Thus fundamental groups of high-dimensional manifolds can be extremely complicated.

1.8.2 Good news: the Whitney trick

The Whitney trick removes pairs of transverse intersection points with opposite local signs. Suppose embedded submanifolds K and L meet transversely at points p and q. Choose arcs

$$\alpha \subset K, \qquad \beta \subset L$$

joining p to q. Their union is the *Whitney circle*. If this circle bounds a framed embedded disk D whose interior is disjoint from $K \cup L$, then an isotopy supported near D eliminates the pair p, q.

In the simply connected high-dimensional situations arising in the proof of the h-cobordism theorem, null-homotopy of the Whitney circle and general-position arguments provide the required Whitney disks. This is the geometric reason that intersection-cancellation methods become effective in dimensions at least five.

Reserved illustration

Whitney trick: transverse intersections of K and L, the arcs α, β, and a Whitney disk.

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Figure 4: Whitney trick: transverse intersections of K and L, the arcs α, β, and a Whitney disk.

1.9 Some classical homotopy-theoretic results

Theorem 1.9 (Hurewicz)**.** Let X be a CW complex. Suppose

$$\pi_i(X) = 0 \qquad \text{for every } i < n,$$

where $n \geq 2$. Equivalently, X is $(n-1)$-connected. Then

$$H_i(X) = 0 \qquad \text{for every } i < n,$$

and the Hurewicz homomorphism

$$\pi_n(X) \longrightarrow H_n(X), \qquad [S^n \xrightarrow{f} X] \longmapsto f_*[S^n],$$

is an isomorphism.

Theorem 1.10 (Whitehead)**.** Let X and Y be CW complexes and let $f : X \to Y$ be continuous. If

$$f_* : \pi_i(X) \xrightarrow{\cong} \pi_i(Y)$$

for all homotopy groups, then f is a homotopy equivalence.

Corollary 1.11. Let X and Y be simply connected CW complexes. If a continuous map

$$f : X \longrightarrow Y$$

induces isomorphisms

$$f_* : H_i(X) \xrightarrow{\cong} H_i(Y)$$

for all i, then f is a homotopy equivalence.

This corollary is often called the homological Whitehead theorem in the simply connected case.

2 The h-Cobordism Theorem and Applications

2.1 Cobordisms and h-cobordisms

Definition 2.1. A *cobordism* W is a compact manifold with boundary together with a decomposition

$$\partial W = \partial^+ W \sqcup \partial^- W.$$

Reserved illustration

Cobordism W with boundary decomposition $\partial W = \partial^+ W \sqcup \partial^- W$.

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Figure 5: Cobordism W with boundary decomposition $\partial W = \partial^+ W \sqcup \partial^- W$.

Definition 2.2. A cobordism $(W; \partial^+ W, \partial^- W)$ is an *h-cobordism* if both inclusions

$$\partial^+ W \hookrightarrow W, \qquad \partial^- W \hookrightarrow W$$

are homotopy equivalences.

2.2 Smale's h-cobordism theorem

Theorem 2.3 (h-cobordism theorem, Smale)**.** Let W^n be a smooth simply connected h-cobordism with $n \geq 6$. Then there exists a diffeomorphism

$$F : \partial^- W \times [0,1] \xrightarrow{\cong} W$$

which is the identity on $\partial^- W \times \{0\}$ (after the usual identification). In particular, the two boundary components are diffeomorphic.

2.3 Application to the generalized Poincaré conjecture

Let M^n be a closed smooth manifold with $n \geq 6$, and assume

$$\pi_1(M) = 0, \qquad H_*(M) \cong H_*(S^n).$$

Choose two disjoint embedded n-balls $D^n_+, D^n_- \subset M$ and set

$$W^n = M^n \setminus \left(\operatorname{int} D^n_+ \cup \operatorname{int} D^n_-\right).$$

Then

$$\partial^+ W \cong S^{n-1}, \qquad \partial^- W \cong S^{n-1}.$$

The goal is to show that $(W; \partial^+ W, \partial^- W)$ is an h-cobordism.

2.3.1 Fact 1: W is simply connected

Since $n \geq 6$, removing finitely many open n-balls does not change the fundamental group. Van Kampen's theorem therefore gives

$$\pi_1(W) = 0.$$

Reserved illustration

Generalized Poincaré conjecture setup: remove two open n-balls from M^n to obtain W with $\partial^{\pm}W \cong S^{n-1}$.

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Figure 6: Generalized Poincaré conjecture setup: remove two open n-balls from M^n to obtain W with $\partial^{\pm}W \cong S^{n-1}$.

2.3.2 Fact 2: the boundary inclusions are homology equivalences

Let

$$i_{\pm} : \partial^{\pm}W \hookrightarrow W$$

be the inclusion maps. Set

$$X_{-} = M \setminus \operatorname{int} D^n_{-}.$$

Because M has the integral homology of S^n, the manifold X_- has the homology of a disk. Moreover,

$$W = X_{-} \setminus \operatorname{int} D^n_{+}.$$

By excision,

$$H_*(X_-, W) \cong H_*(D^n_+, S^{n-1}).$$

The long exact sequence of the pair (X_-, W) then shows that

$$H_i(W) \cong \begin{cases} \mathbb{Z}, & i = 0, n-1, \\ 0, & \text{otherwise}, \end{cases}$$

and that the inclusion $\partial^+ W \cong S^{n-1} \hookrightarrow W$ induces an isomorphism on homology. Repeating the same argument with $+$ and $-$ interchanged proves the same statement for $\partial^- W$.

Since W and both boundary spheres are simply connected CW complexes, the homological Whitehead theorem implies that both inclusions are homotopy equivalences. Hence $(W; \partial^+ W, \partial^- W)$ is an h-cobordism.

2.4 From the h-cobordism theorem to a sphere

By the h-cobordism theorem,

$$W \cong \partial^- W \times [0,1] \cong S^{n-1} \times [0,1].$$

Thus, after reattaching the two n-balls, the manifold M is obtained by gluing two copies of D^n along a self-diffeomorphism of S^{n-1}.

Reserved illustration

From the h-cobordism theorem to a sphere: $W \cong S^{n-1} \times [0,1]$ and reattaching the two n-balls.

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Figure 7: From the h-cobordism theorem to a sphere: $W \cong S^{n-1} \times [0,1]$ and reattaching the two n-balls.

2.5 The Alexander trick

Theorem 2.4 (Alexander trick)**.** Let

$$f : S^{n-1} \longrightarrow S^{n-1}$$

be a homeomorphism (in particular, a diffeomorphism). Then f extends to a homeomorphism

$$C(f) : D^n \longrightarrow D^n$$

given by

$$C(f)(x) = \begin{cases} \|x\| \, f\left(\dfrac{x}{\|x\|}\right), & x \neq 0, \\ 0, & x = 0. \end{cases}$$

Geometrically, this extends the boundary map radially along each line segment from the origin.

Reserved illustration

Alexander trick: radial extension of $f : S^{n-1} \to S^{n-1}$ to $C(f) : D^n \to D^n$.

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Figure 8: Alexander trick: radial extension of $f : S^{n-1} \to S^{n-1}$ to $C(f) : D^n \to D^n$.

Combining the h-cobordism theorem with the Alexander trick yields a homeomorphism

$$M^n \cong_{\text{homeo}} S^n \qquad (n \geq 6)$$

in the setting above.

2.6 Exotic spheres and gluing two disks

Corollary 2.5. Let Σ^n be an exotic sphere in dimension $n \geq 6$, so in particular

$$\Sigma^n \cong_{\text{homeo}} S^n.$$

Then Σ^n can be written as

$$\Sigma^n \cong D^n \cup_f D^n,$$

where

$$f : S^{n-1} \longrightarrow S^{n-1}$$

is a diffeomorphism used as the gluing map.

Summary

Homotopy type alone does not determine a manifold: lens spaces and rational Pontryagin classes provide basic obstructions. Aspherical manifolds exhibit a contrasting rigidity phenomenon, expressed by the Borel conjecture and, in the hyperbolic case, by Mostow rigidity. In high-dimensional simply connected topology, the Whitney trick converts geometric intersection problems into handle cancellation, and Smale's h-cobordism theorem then yields the high-dimensional Poincaré theorem and the disk-gluing description of homotopy spheres.